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Isometría de un sistema ortonormal finito a $\mathbb{K}^n$

Proposición 1. *Sea H un espacio prehilbert y $S = \{u_i\}_{i=1}^n$ un sistema ortonormal*

$$\implies T\colon \operatorname{span}(S) \longrightarrow \mathbb{K}^n \quad \text{es una isometría biyectiva con la métrica euclídea en } \mathbb{K}^n.$$
$$x \longmapsto (\langle x, u_1\rangle, \ldots, \langle x, u_n\rangle)$$

***Demostración*:** Sea $x \in \operatorname{span}\{u_i\}_{i=1}^n$. Entonces, $\exists \{\lambda_i\}_{i=1}^n \subset \mathbb{K}$ tal que $x = \sum_{i=1}^n \lambda_i u_i$. Pero como S es un sistema ortonormal, se tiene que

$$\forall i \in \mathbb{N}_n : \langle x, u_i\rangle = \left\langle \sum_{j=1}^n \lambda_j u_j, u_i \right\rangle = \sum_{j=1}^n \lambda_j \langle u_j, u_i\rangle = \sum_{j=1}^n \lambda_j \delta_{ij} = \lambda_i.$$

Por tanto, $x = \sum_{i=1}^n \langle x, u_i\rangle u_i$, luego

$$\|x\|^2 = \langle x, x\rangle = \left\langle \sum_{i=1}^n \langle x, u_i\rangle u_i, x \right\rangle = \sum_{i=1}^n \langle x, u_i\rangle \overline{\langle x, u_i\rangle} = \sum_{i=1}^n |\langle x, u_i\rangle|^2 = \|T(x)\|_{\mathbb{K}^n}^2 .$$

Veamos ahora que T es biyectiva. Para ello, como es isometría, basta ver que es sobreyectiva. Sea $(\alpha_1, \ldots, \alpha_n) \in \mathbb{K}^n$. Consideramos $x = \sum_{i=1}^n \alpha_i u_i \in \operatorname{span}(S)$. Entonces,

$$T(x) = (\langle x, u_1\rangle, \ldots, \langle x, u_n\rangle) = (\alpha_1, \ldots, \alpha_n).$$

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Referenciado en

- `prop-subesp-vectorial-generado-cerrado`

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